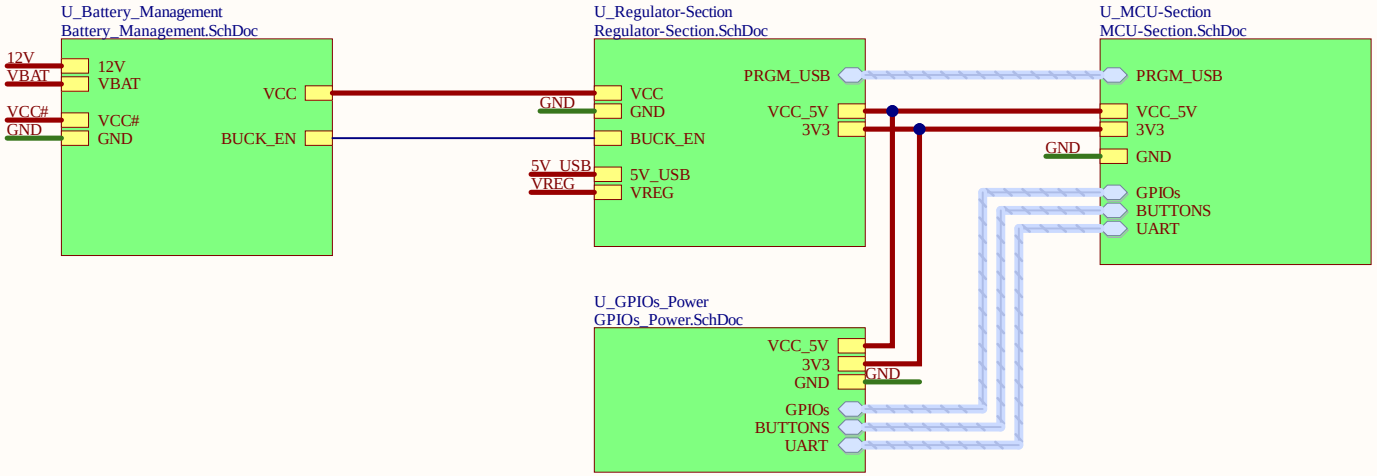
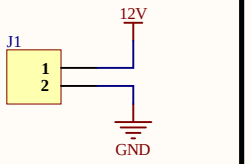


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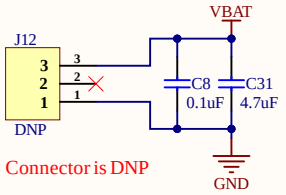
Title: Top_Sheet.SchDoc	
File: Master.PrjPcb	
Date: 7/19/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 5

CHARGING PORT



Charger rail should be capable of 12V / 3A ratings

BATTERY INPUT

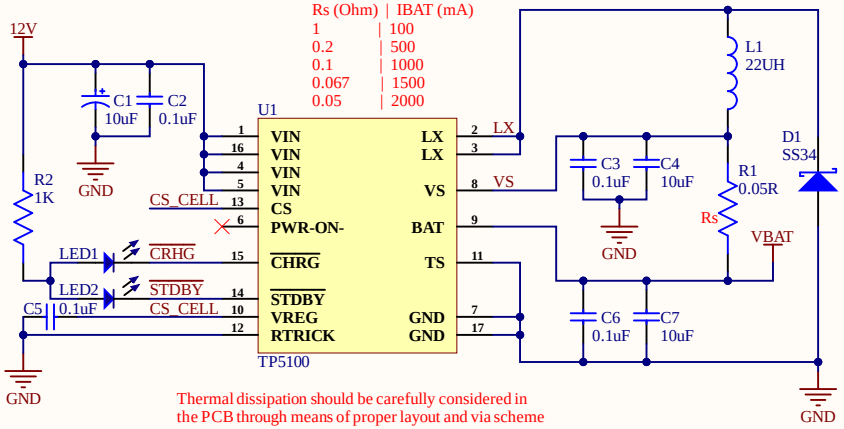


Connector is DNP

Battery Capacity 7.4V 9800mA
Decoulping is added for better stability

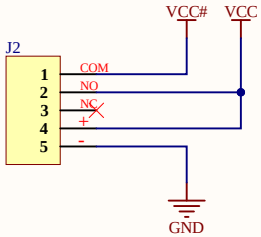
BATTERY CHARGER WIH POWER PRIORITY

Rs (Ohm)	IBAT (mA)
1	100
0.2	500
0.1	1000
0.067	1500
0.05	2000



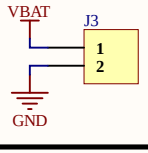
Thermal dissipation should be carefully considered in the PCB through means of proper layout and via scheme

LATCH SWITCH

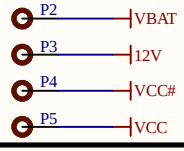


Naming of Pads should be done like COM NO NC + -
VCC will go in main circuit

BATTERY MONITORING



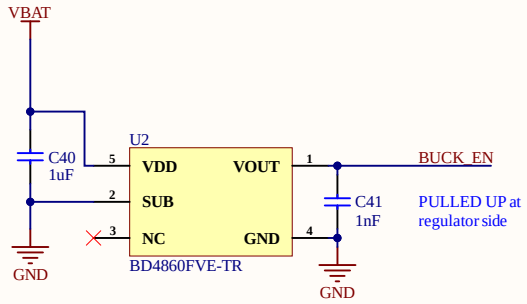
TEST POINTS





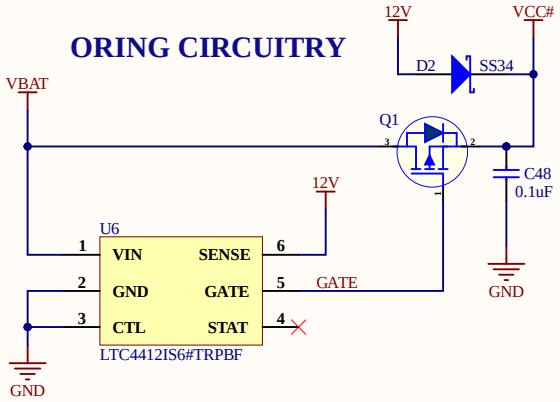


BATTERY SUPERVISORY CIRCUIT



Active LOW. When battery voltage drops below 6V, VOUT will become LOW disabling power section

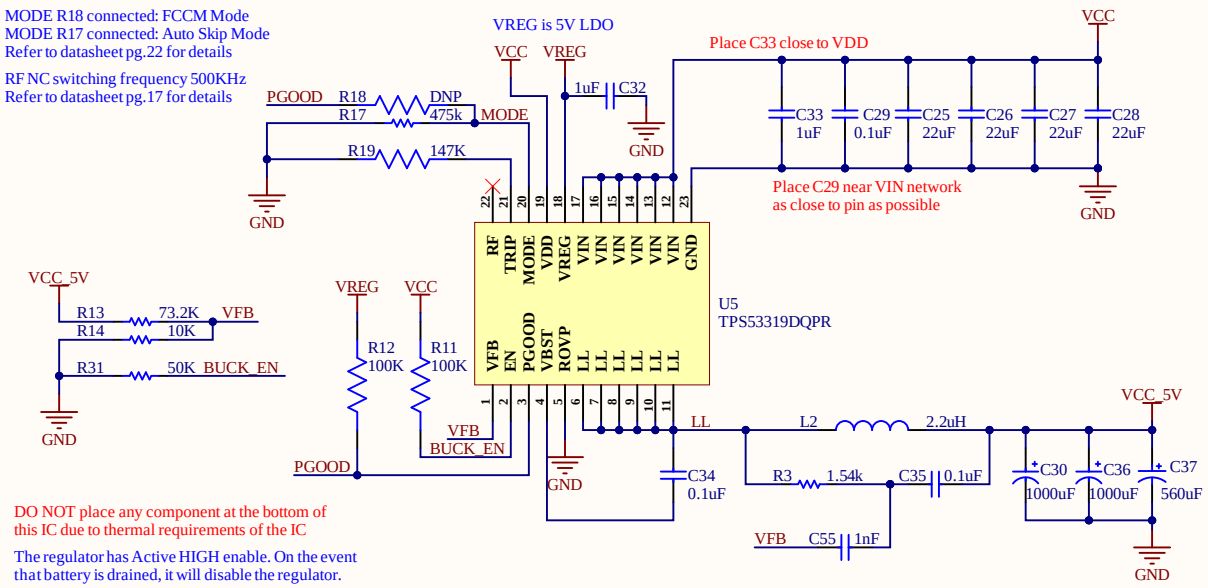
ORING CIRCUITRY



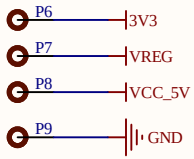
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D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Battery_Management.SchDoc	
File: Master.PrjPcb	
Date: 7/19/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 5

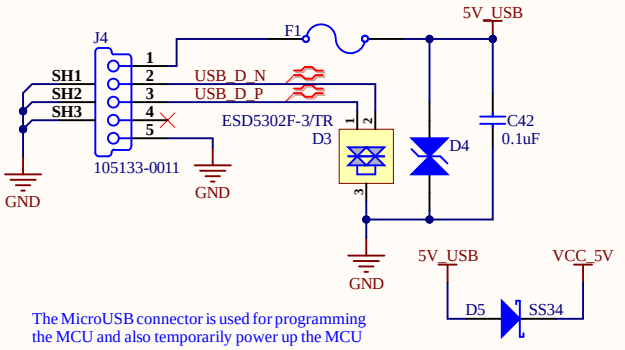
MAIN BUCK REGULATOR CIRCUIT 7.4V / 12V - 5V CONVERSION



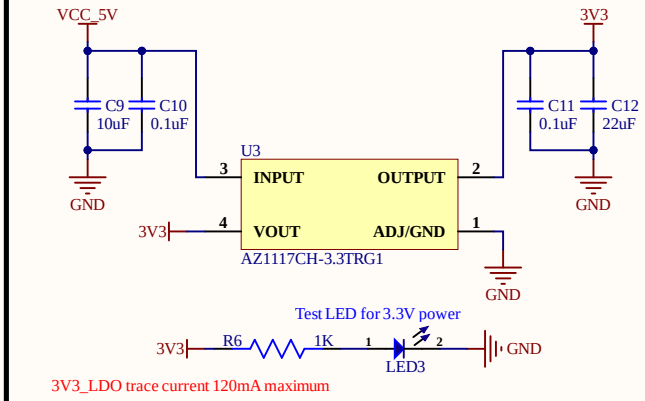
TEST POINTS

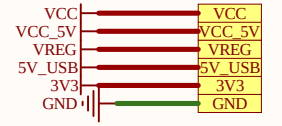


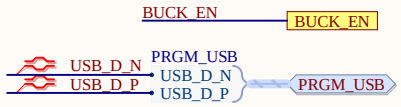
USB INPUT FOR PROGRAMMING



MCU POWER REGULATOR



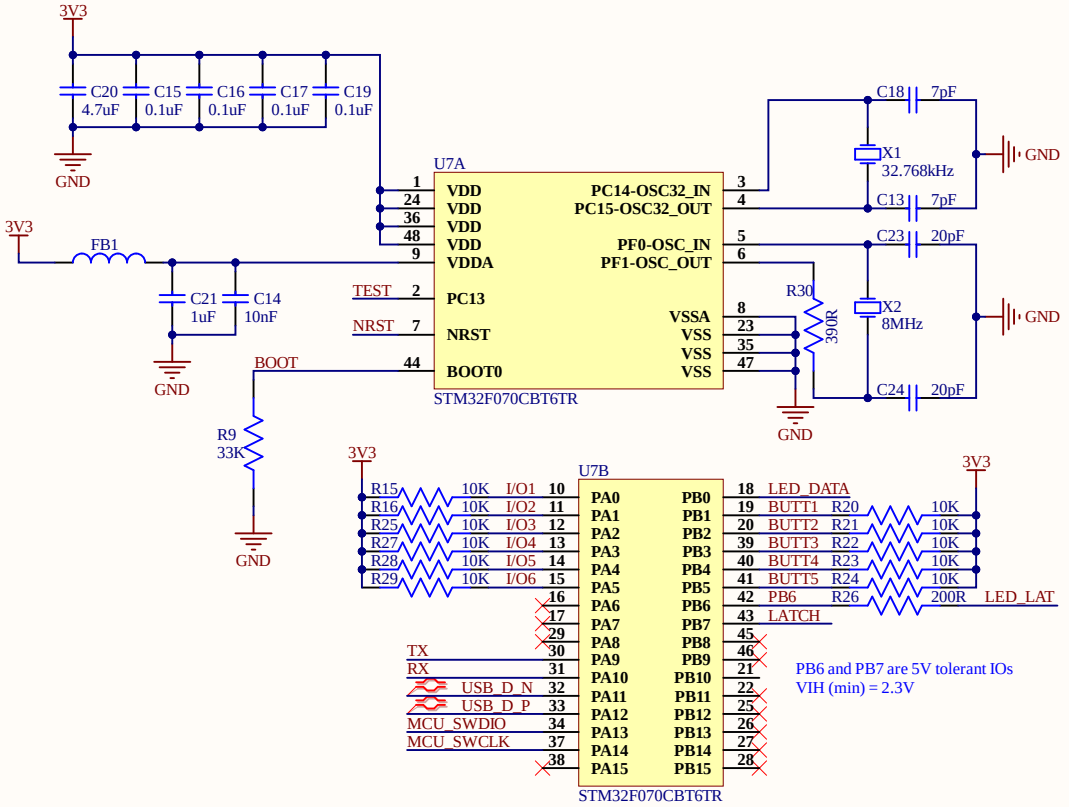




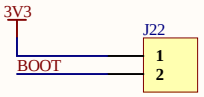
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Regulator-Section.SchDoc	
File: Master.PrjPcb	
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MICROCONTROLLER POWER AND LOGIC CIRCUIT

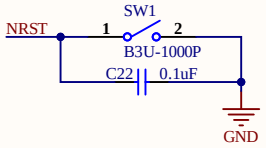


BOOT



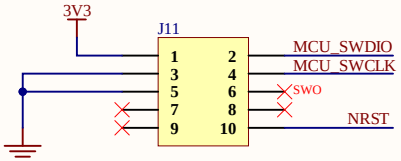
To program MCU, short these 2 pins via Jumper

RESET



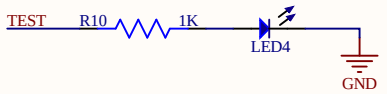
NRST is active LOW and is pulled up internally

DEBUGGING PINS

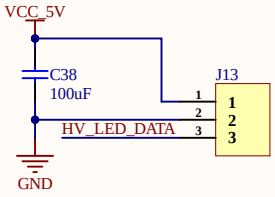


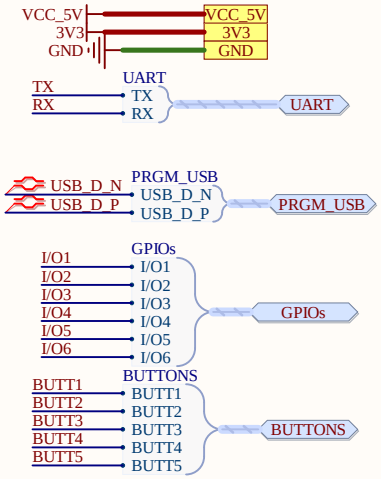
SWO feature is not supported on this Controller

TESTING LED

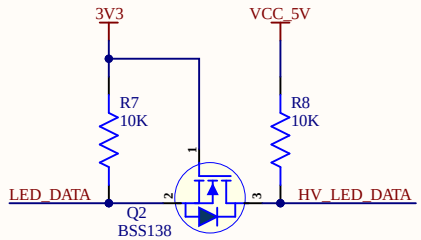


HOT-SWAPPABLE CONNECTION



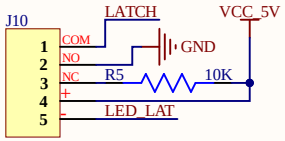


LEVEL SHIFTER



Bi-directional level shifter for WS2812B LED from 3.3V to 5V

LED LATCHING

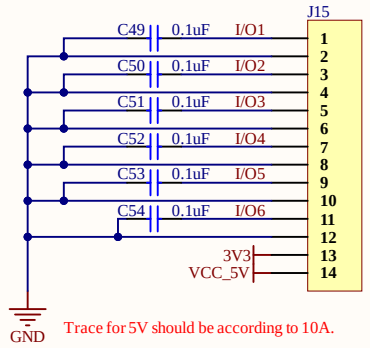


Naming of Pads should be done like COM NO NC + -

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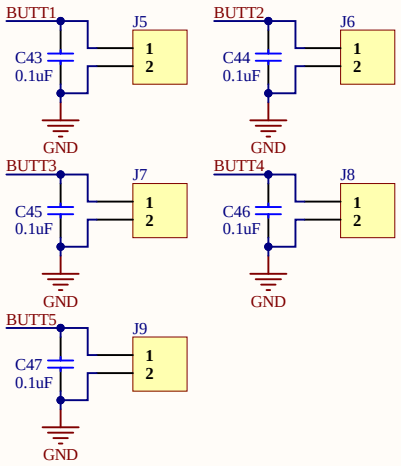
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File: Master.PrjPcb	
Date: 7/19/2023	Page Num: 4
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GPIOs

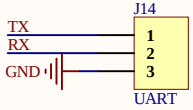


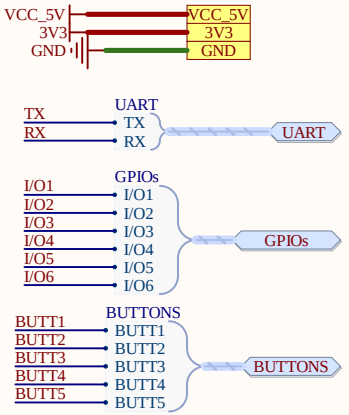
Trace for 5V should be according to 10A.

PUSH BUTTONS



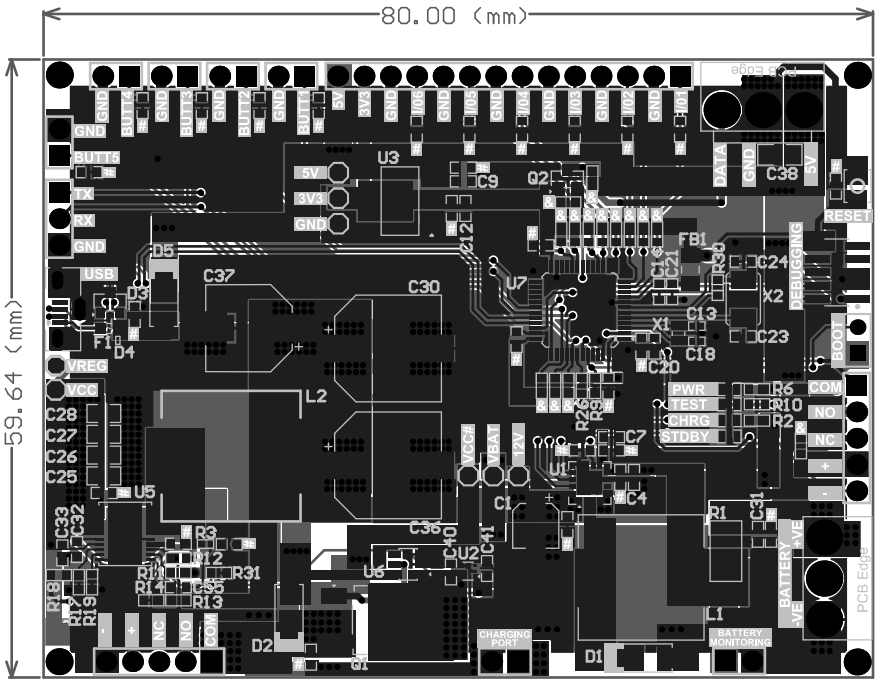
UART





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Title: GPIOs_Power.SchDoc	
File: Master.PrjPcb	
Date: 7/19/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 5



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	3.00mil	3.5	
1	Top Layer		10.00mil		
	Dielectric 1	FR-4	37.00mil	4.8	
2	Bottom Layer		10.00mil		
	Bottom Solder	Solder Resist	3.00mil	3.5	
	Bottom Overlay				
Total board thickness:			63.00mil		